REPUBLIC POLYTECHNIC

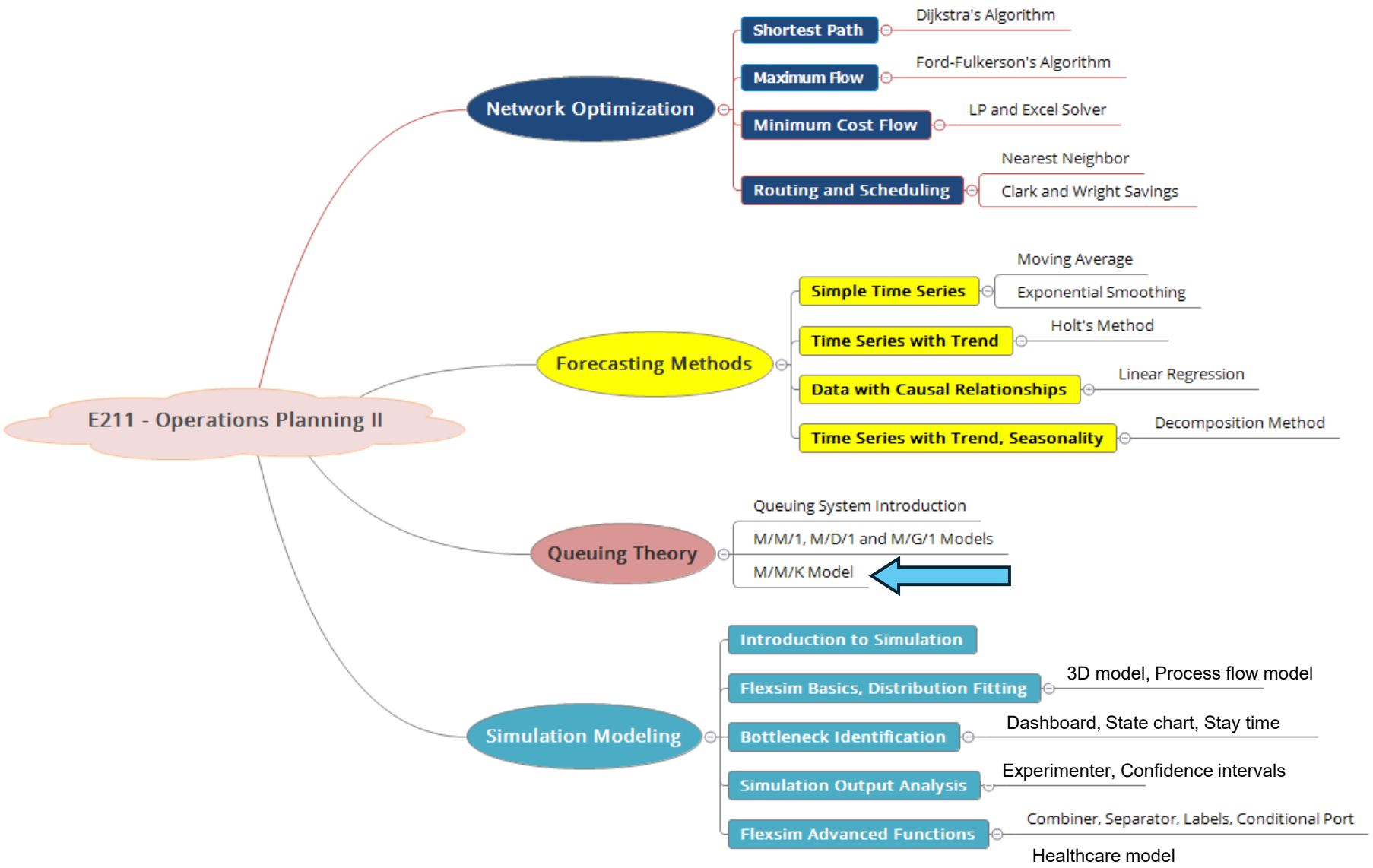
Lesson 3

Queuing Theory – Multiple Server Model (M/M/k)

E211 – Operations Planning II

REPUBLIC POLYTECHNIC

E211 Topic Tree



Recap on Single Server Queuing Models

Kendall's Notation

A/B/C

Arrival Pattern

Service Pattern

Number of Servers

What does M/M/1 mean?

What about M/G/1? M/D/1?

Single Server Model Equations

	M/M/1	M/G/1	M/D/1
Arrival process	Poisson	Poisson	Poisson
Service time distribution	Exponential	General distribution with standard deviation	Deterministic (constant)
Utilization, ρ	Arrival rate / service rate or λ / μ		
Average number in queue, Lq	$Lq = \rho^2 / (1 - \rho)$	$Lq = (\lambda^2 \sigma^2 + \rho^2) / (2(1 - \rho))$	$Lq = \rho^2 / (2(1 - \rho))$
Average number in system, Ls	$Ls = Lq + \lambda / \mu$		
Average time in queue, Wq	$Wq = Lq / \lambda$		
Average time in system, Ws	$Ws = Wq + 1 / \mu$		

Build Your Excel Queuing Model

At a convenience store, a cashier take on average 48 seconds to process a checkout and on average 63 customers arrive every hour.

Your task is to calculate the performance measures for this system under three different models: M/M/1, M/G/1, and M/D/1.

Queuing Models

M/M/k Multi-server Model

Multi-server (M/M/k) Model Characteristics

- Input source: Infinite population; no balks, no reneging
- Arrival process: Number of arrivals follows **Poisson distribution**
- Queue: Unlimited; single line
- Queue discipline: FIFO (FCFS)
- Service mechanism: **Multiple servers,** single phase system
- Service time: **Negative exponential distribution**
- Arrivals and service times are independent of the number of customers in system
- Arrival rate must be smaller than sum of service rate of servers, i.e.

  $\lambda < K\mu$ **(for K servers)**

Little's formula

- Little's Formulas represent important relationships between L_s, L_q, W_s and W_q.
- Conditions:
  - System has Single Queue,
  - Customers arrive at a finite arrival rate λ, and
  - System operates in steady state
- No assumptions on arrival or service time distributions or queue limits. The formulae are applicable to M/M/1, M/G/1, M/D/1 and M/M/k queuing systems:

Average number in system, $L_s = L_q + \lambda / \mu$

Average time in queue, $W_q = \frac{L_q}{\lambda}$

Average time in system, $W_s = W_q + 1 / \mu$

M/M/k Performance Measures

MULTIPLE SERVER QUEUE: M/M/k

k = number of channels or servers

Utilization Factor = $\lambda / k\mu$

$$P_0 = \frac{1}{\left[\sum_{n=0}^{k-1} \frac{(\lambda/\mu)^n}{n!}\right] + \frac{(\lambda/\mu)^k}{k!}\left(\frac{k\mu}{k\mu - \lambda}\right)}$$

For example if k = 3: $P_0 = \dfrac{1}{1 + \frac{\lambda}{\mu} + \frac{(\lambda/\mu)^2}{2!} + \frac{(\lambda/\mu)^3}{3!}\left(\frac{3\mu}{3\mu}\right)}$

Continue in this order:

L_s

$$L_q = \frac{(\lambda/\mu)^k \lambda\mu}{(k-1)!(k\mu - \lambda)^2} P_0$$

$$L = L_q + \frac{\lambda}{\mu}$$

W_s

$$W_q = \frac{L_q}{\lambda}$$

$$W = W_q + \frac{1}{\mu}$$

$$P_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{k!\,k^{n-k}} P_0 & \text{for } n > k \\ \dfrac{(\lambda/\mu)^n}{n!} P_0 & \text{for } 0 \le n \le k \end{cases}$$

$$P_w = \frac{1}{k!}\left(\frac{\lambda}{\mu}\right)^k \left(\frac{k\mu}{k\mu - \lambda}\right) P_0$$

P_0 =probability of no object in the queuing system

P_n =probability of n objects in the queuing system

P_w =probability of at least k objects in the queuing system; i.e., probability that an object needs to wait before getting the service.

Exercise Scenario

- People arrive at a security checkpoint at a rate of 1 person every 30 seconds. The security check takes 5 minutes on average. There are 12 security counters in operation. Calculate the performance measures of the M/M/k queuing system.

- State the queuing model using Kendall's notation

M/M/12

- What is the Arrival Rate, λ?

λ = 60/30 = 2 per min

- What is the Service Rate, μ?

μ = 1/5 per minute = 0.2 per min

- What is the System Utilization, ρ?

$\rho = \lambda/(k\mu) = 2 / (12x(1/5)) = 0.833$

Exercise Scenario

- People arrive at a security checkpoint at a rate of 1 person every 30 seconds. The security check takes 5 minutes on average. There are 12 security counters in operation. Calculate the performance measures of the M/M/k queuing system.

- **Probability of no customer, $P_0 = 3.59 \times 10^{-05}$**

- **Number in queue, Lq = 2.25**

- **Number in system, Ls = Lq + λ/μ = 2.247 + 2/ (1/5) = 12.25**

- **Time in queue, Wq = Lq/ λ = 2.247/ 2 = 1.12 mins**

- **Time in system, Ws = Wq + 1/ μ = 1.124 + 5 = 6.12 mins**

Build Your Excel Queuing Model

Use the Excel provided previously to build the M/M/k queuing model.

Answer these questions:

1. How will the waiting time change if one of the 12 counters is closed?
2. What is the minimum number of counters needed?
3. What happens if there are 15 counters?
4. What is the probability that at least 2 of the counters will be idle?
5. What is the probability that there will be 6 or more people in the queue?

Build Your Excel Queuing Model (Answer)

Use the Excel provided previously to build the M/M/k queuing model.

Answer these questions:

1. How will the waiting time change if one of the 12 counters is closed? **k = 11. Waiting time increases from 1.12 mins to 3.41 mins**
2. What is the minimum number of counters needed? **k> λ/μ > 10 = 11**
3. What happens if there are 15 counters? **k = 15. Waiting time is reduced to 0.1 min**
4. What is the probability that at least 2 of the counters will be idle? **P(n<=10) = 0.46**
5. What is the probability that there will be 5 or more people in the queue? **P(n>=17) = 1 – P(n<=16) = 0.18**

Application of Queuing Models (I)

1. Analyse Service Operations

Queue length - Long lines indicate poor customer service, inefficient service, or inadequate capacity.

Number of customers in system - A large number causes congestion and dissatisfaction.

Waiting time in line - Long waiting times are associated with poor service.

Total time in system – Long times may indicate problems with customers, server efficiency, or capacity.

Service facility utilization – associated with capacity and costs of providing service

Application of Queuing Models (II)

2. Design Service Operations

Aim is to balance costs against benefits of improving service system.

Arrival rates - Adjusts through advertising, promotions, pricing, appointments.

Number of servers - Determines system capacity.

Number of phases - Consider splitting service into different phases

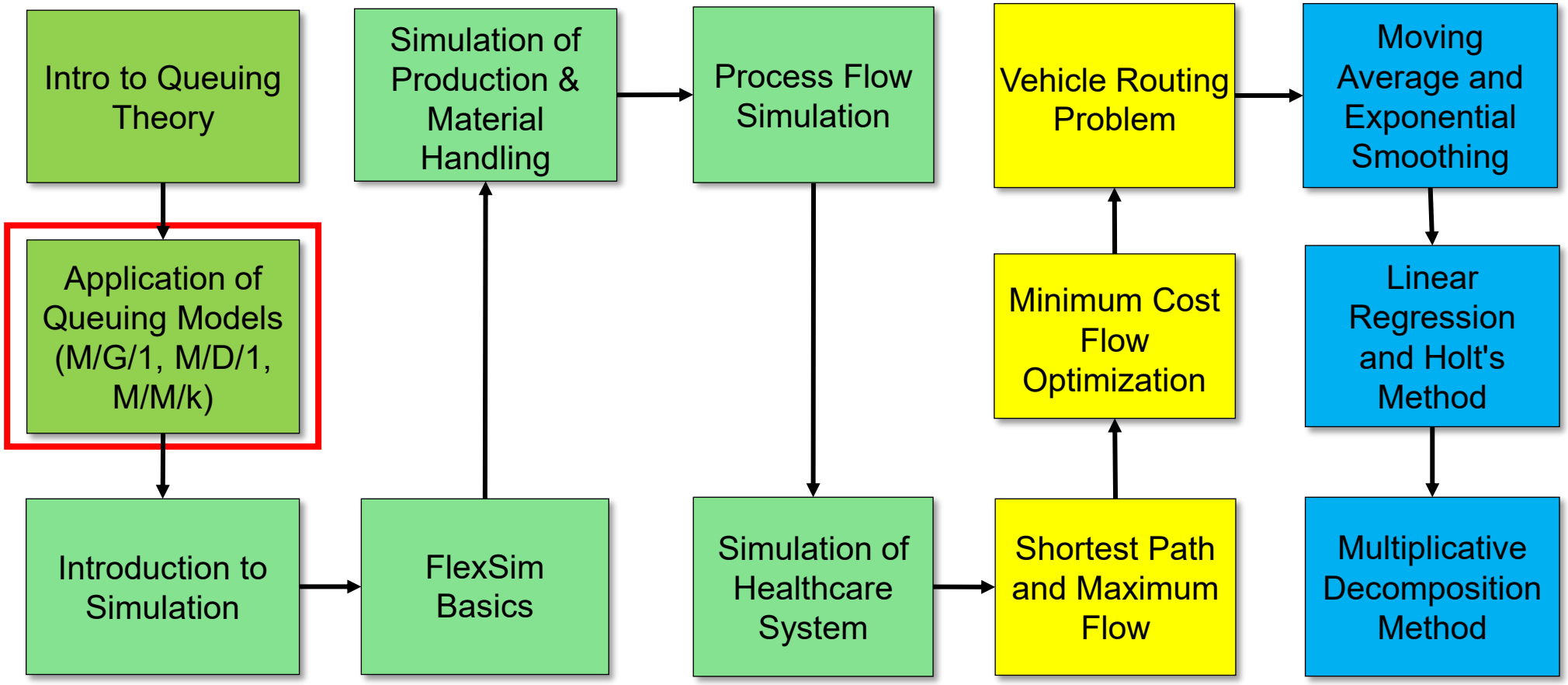
Server efficiency – Determines service time. Improves through training, incentives, work methods and/or capital investment.

Priority rule – Service differentiation for customers.

Line arrangement - Single or multiple lines.

Learning Outcomes

At the end of the lesson, students should be able to:

- Calculate performance measures of M/M/k queuing models.
- Build Excel models of queuing systems.
- Apply queuing models in the design of service operations.

E211 Progression